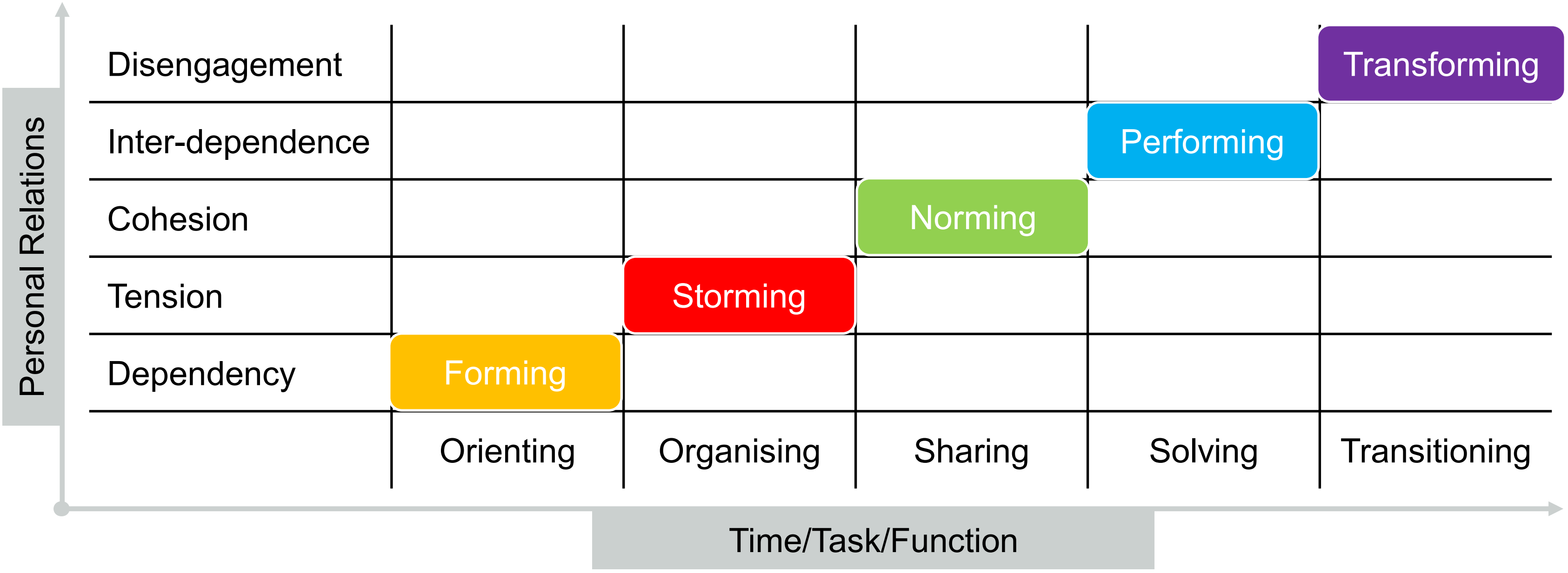
PROJECT MANAGEMENT 2

13th October 2025

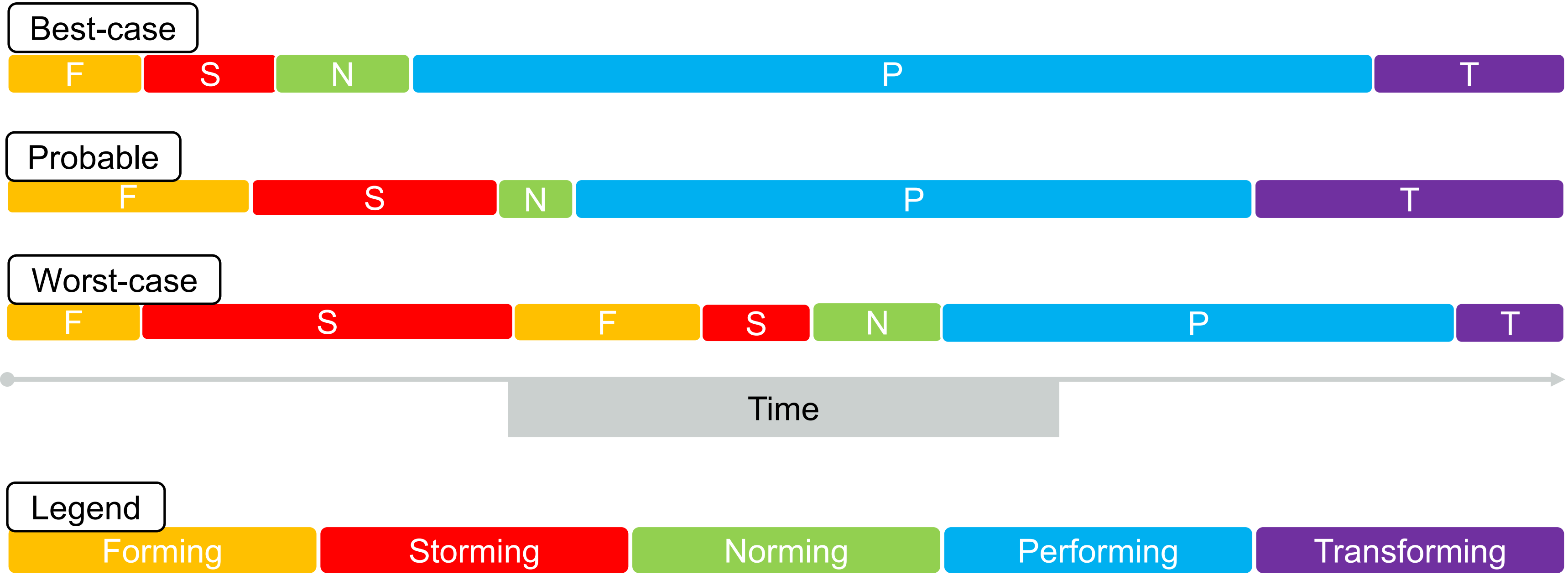
Mark Graham

avdasi2@bristol.ac.uk

The Tuckman model



Scenarios


Best-case
F
S
N
P
T
Probable
F
S
N
P
T
Worst-case
F
S
F
S
N
P
T
Time
Legend
Forming
Storming
Norming
Performing
Transforming

	0 Forming 4	Storming 8	Norming 12	Performing 16
Emotions / Thoughts	-Expectation -Anxiety -Excitement	-Anger -Frustration -Strong concerns about how to meet goals	-More comfort/ease in expression of real feelings -An individual's expectations of the team and reality align	-Satisfaction -Good awareness of strengths and weaknesses of group members
Behaviours	-Shyness -Friendly -Lots of questions (many trivial) -Very polite	-Less polite discussion -Critical language used -Less suppression of feelings & self	-More frequent and meaningful communication between members -Constructive criticism possible, and even welcomed -Team might develop its own culture, jokes, language	-Team able to solve problems -Differences between team members are utilised (and even useful) -Roles can become very fluid now
Team Tasks	-Discussing overall goals -Defining roles -Setting rules and expectations	-Exploring how work actually gets done -Breaking down large goals into smaller ones -Finding problems in the team and systems	-Productivity increases -Useful and detailed plans start to appear -Team begins to be able to assess and monitor its progress	-High levels of team commitment -Progress is recognised and celebrated -Making significant progress on team goals

Project Management - Learning objectives

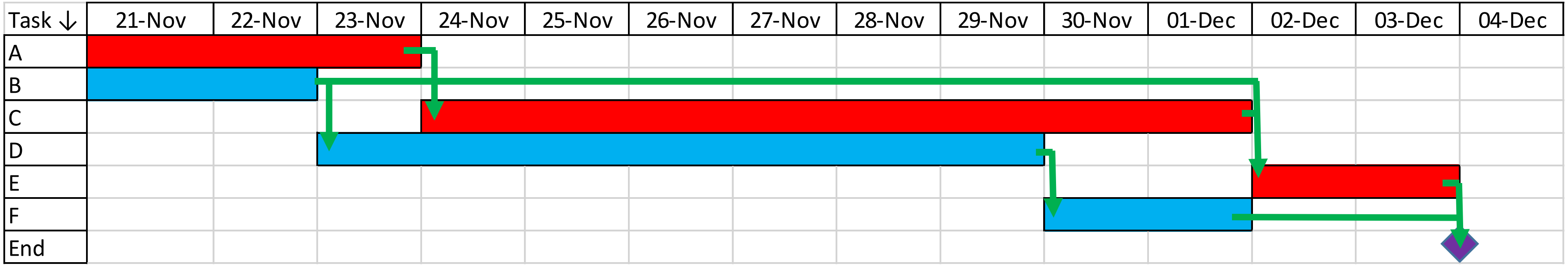
Unit level learning objective(s)

2. work as a member of a team, employing appropriate project management and planning tools to create, monitor and deliver a project plan;

Session level learning objectives

1. **Recall** and **discuss** the attributes of waterfall and agile project management approaches
2. **Create** a work breakdown structure
3. **Identify** task dependencies within a project
4. **Create** a network diagram from a list of task durations and task dependencies
5. **Create** a Gantt chart from a corresponding network diagram (or directly from a list of tasks and task dependencies)
6. **Identify** the critical path of a schedule and the float of any task within a network diagram or Gantt chart
7. **Critically** analyse a project schedule to **identify** risks and opportunities within the plan

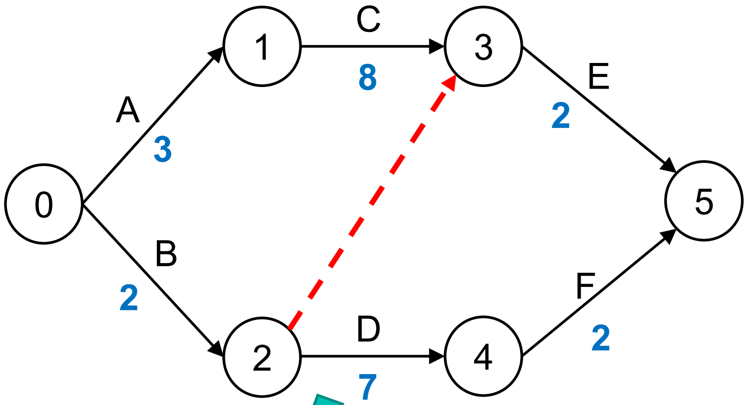
Gantt chart - Principle

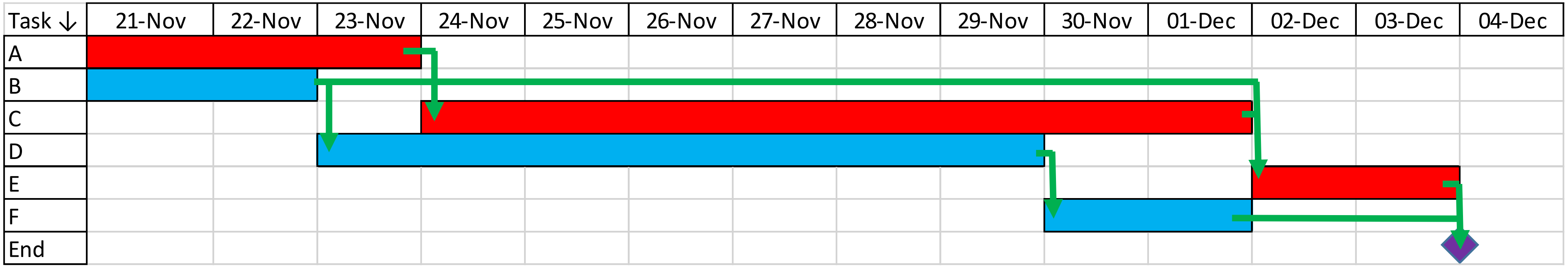


- Horizontal axis = time
- Rows are tasks
- Arranged logically (tasks grouped according to dependency)

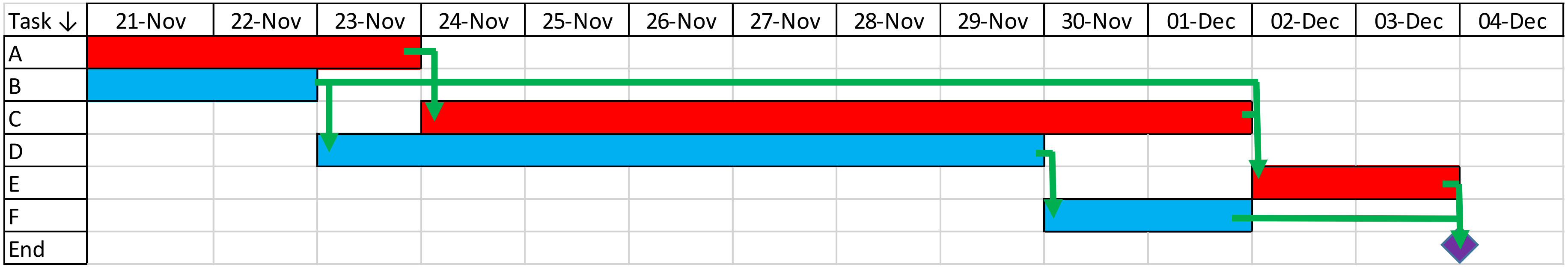
Gantt chart – equivalence with WBS & Network

Activity	Predecessor	Duration (weeks)
A	(start)	3
B	(start)	2
C	A	8
D	B	7
E	B, C	2
F	D	2

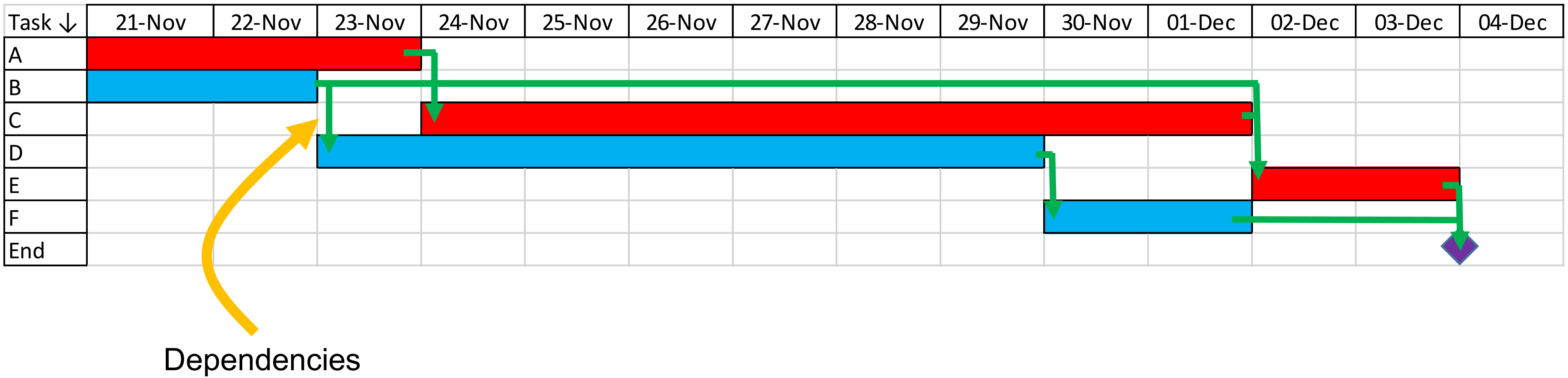




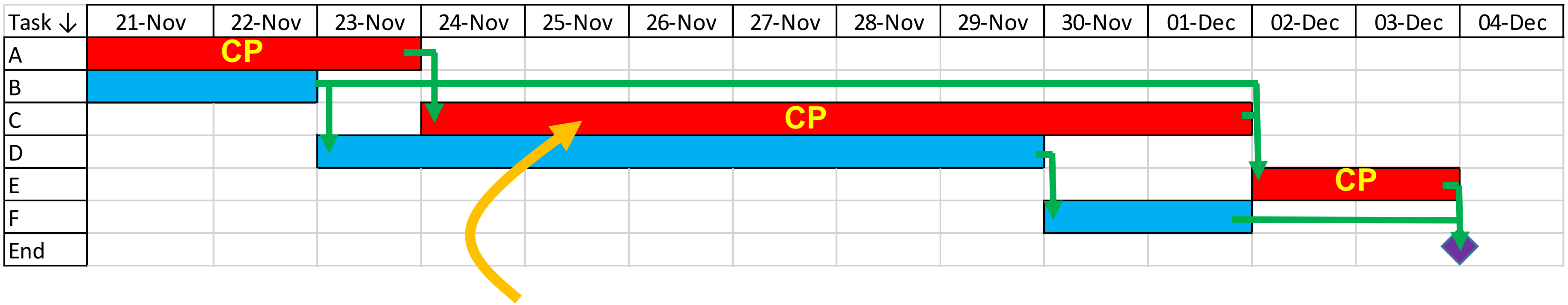
Components of a Gantt chart



Components of a Gantt chart



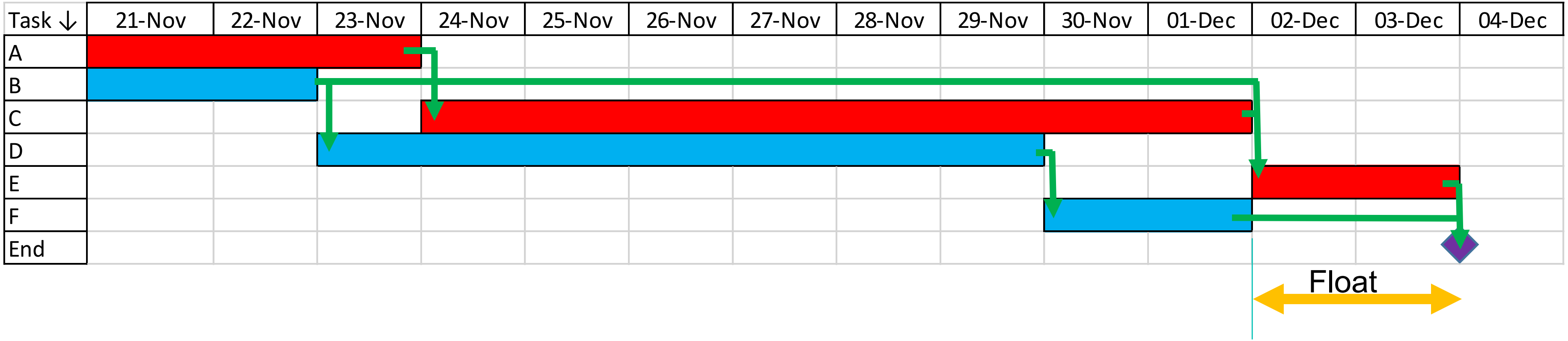
Components of a Gantt chart



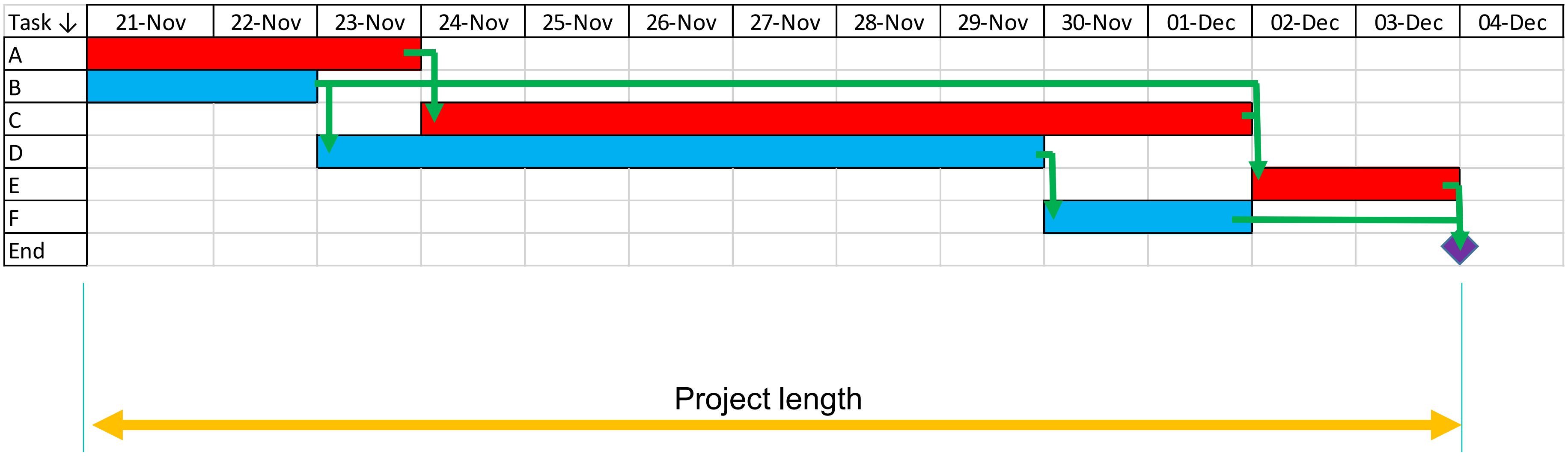
Critical path tasks

Labelled with colour or annotation

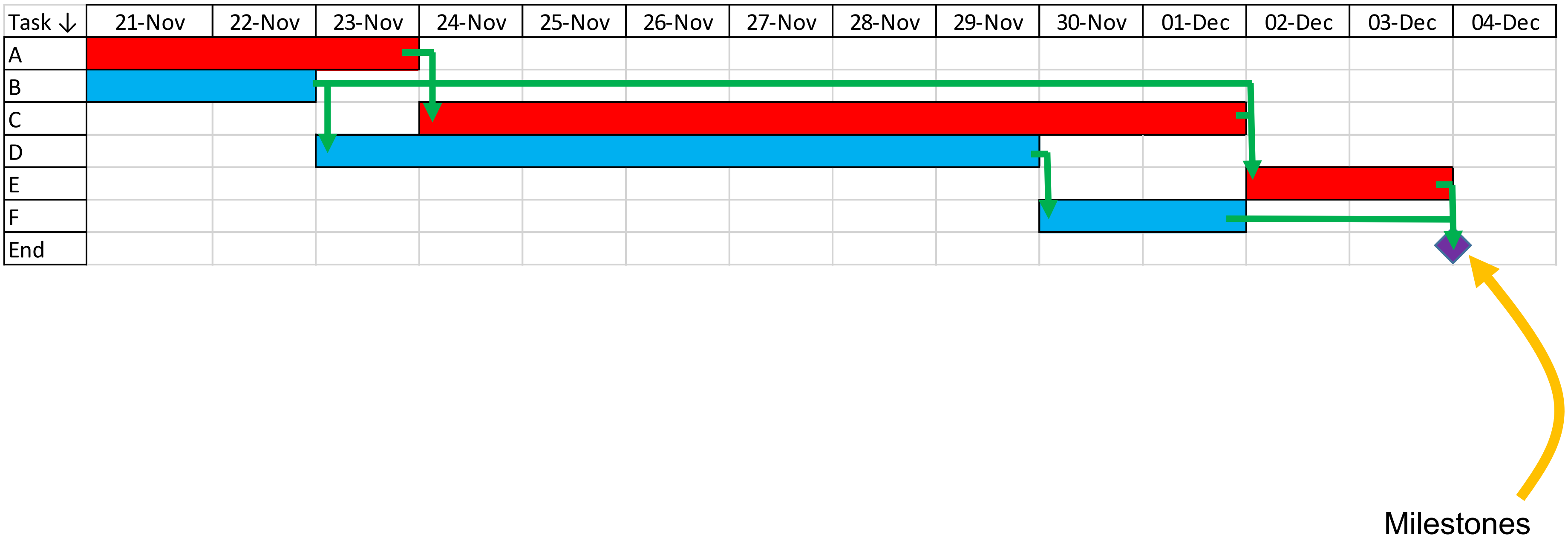
Components of a Gantt chart



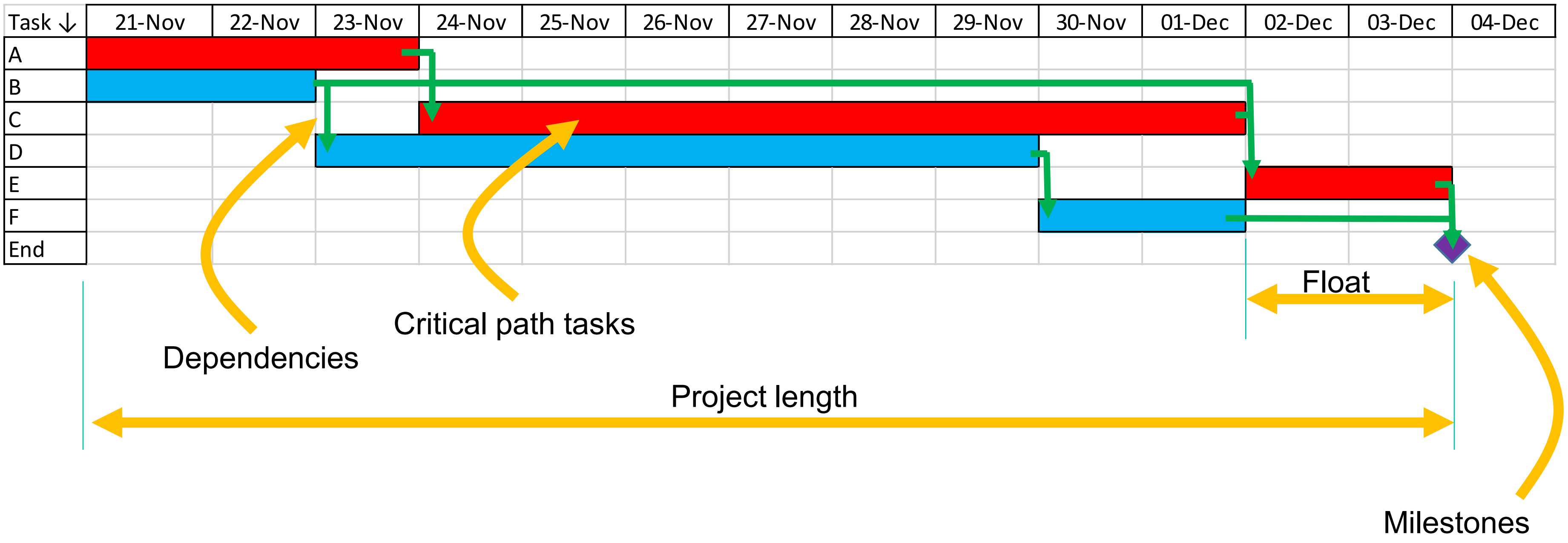
Components of a Gantt chart



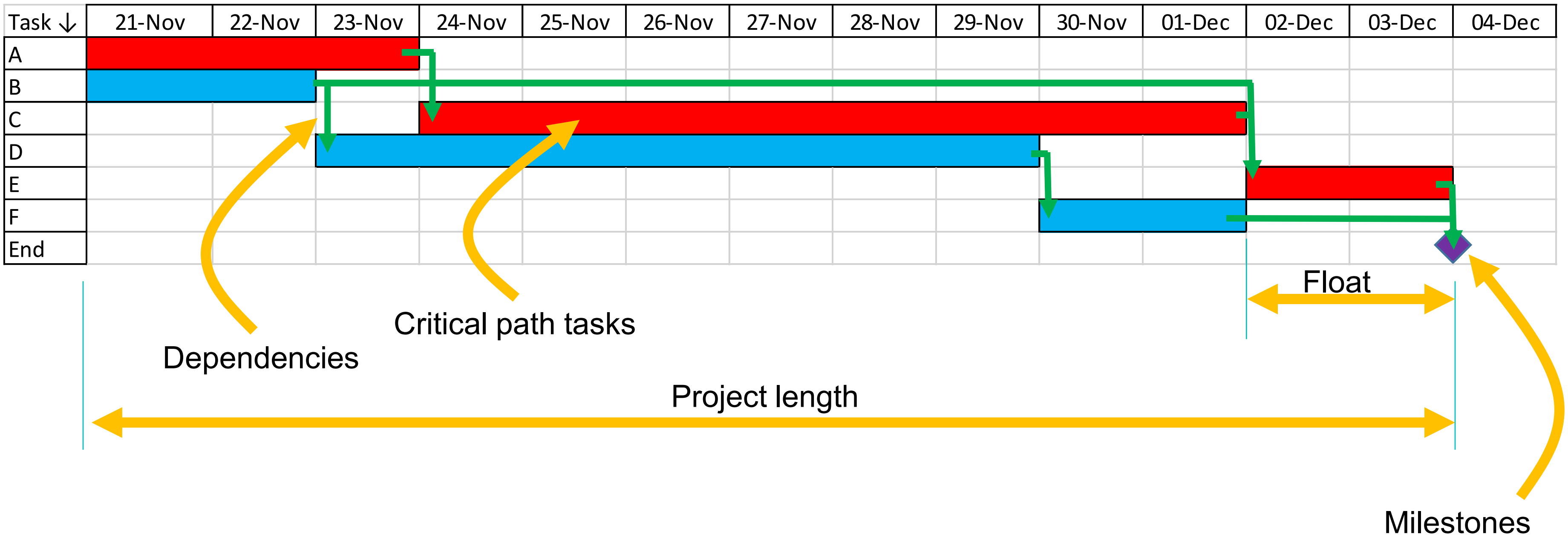
Components of a Gantt chart



Components of a Gantt chart



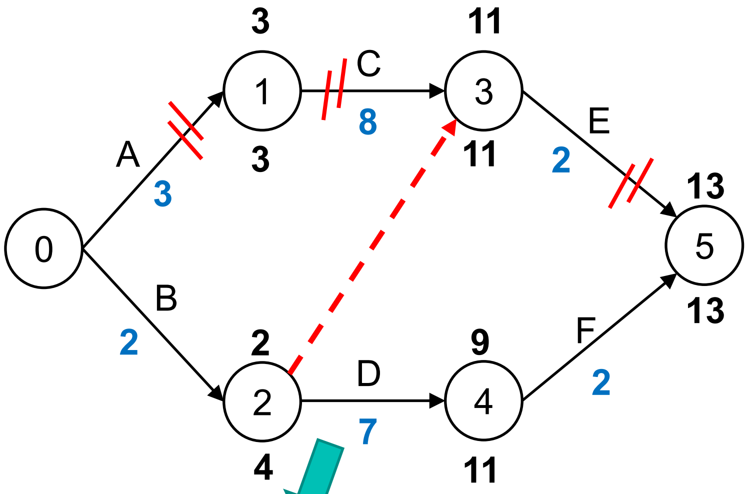
Components of a Gantt chart

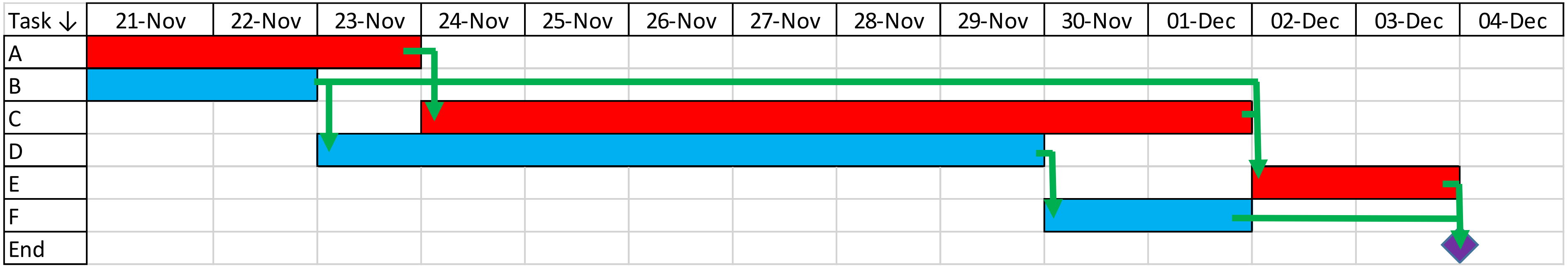


Notice groupings of linked tasks

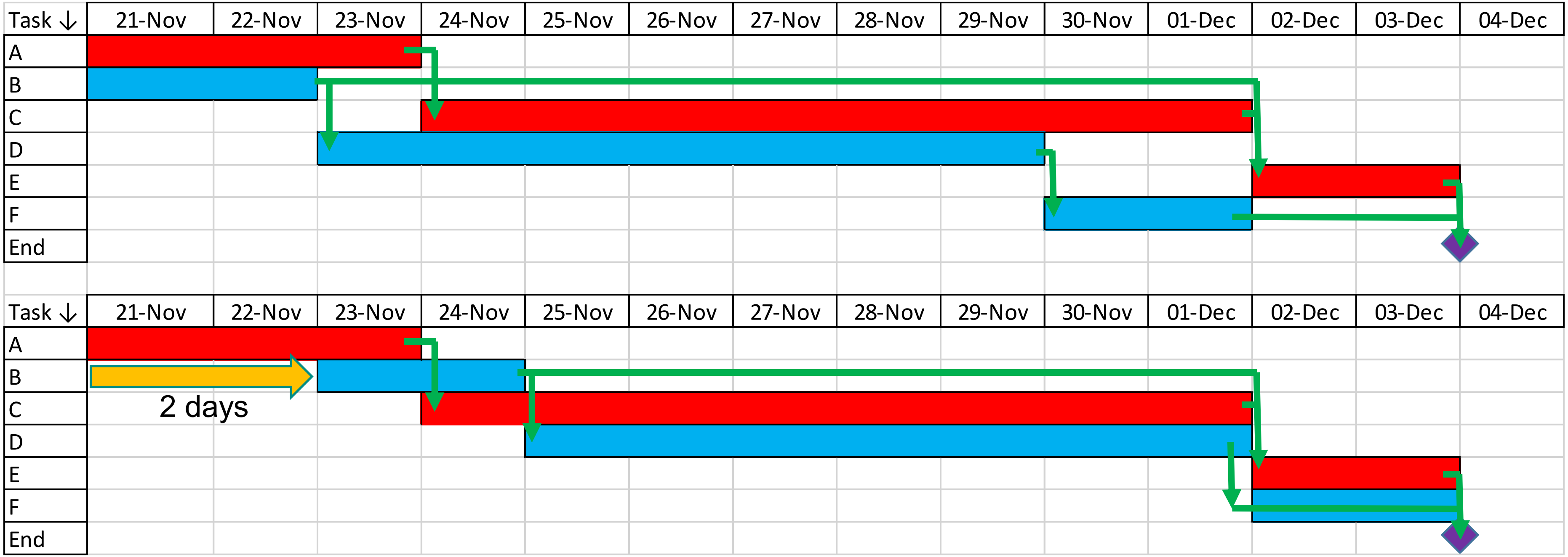
Gantt chart

Activity	Predecessor	Duration (weeks)
A	(start)	3
B	(start)	2
C	A	8
D	B	7
E	B, C	2
F	D	2





Gantt chart – optimising your plan

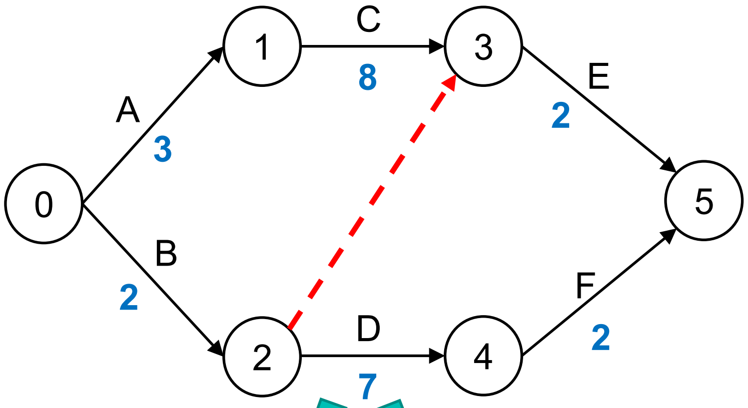


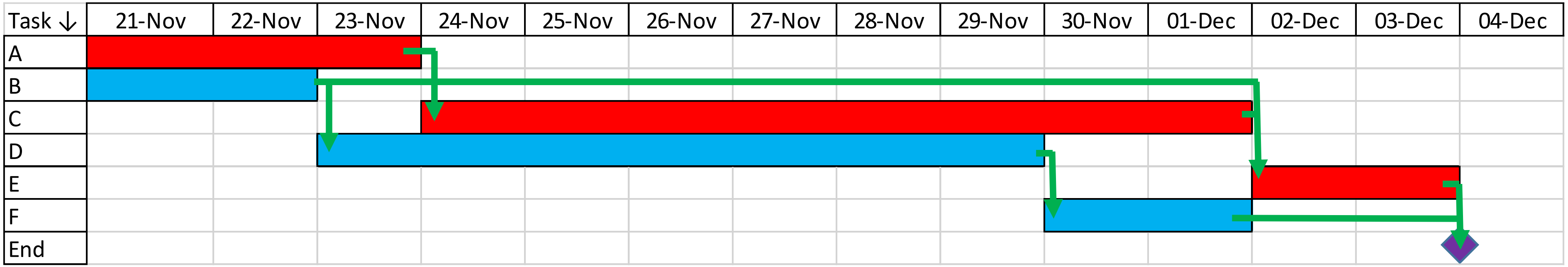
In B → D → F there is a float of 2 days. I could…

- Delay B by 2 days
- Delay B by 1 day and D by 1 day
- and so on…

Equivalence of: Table → Network → Gantt

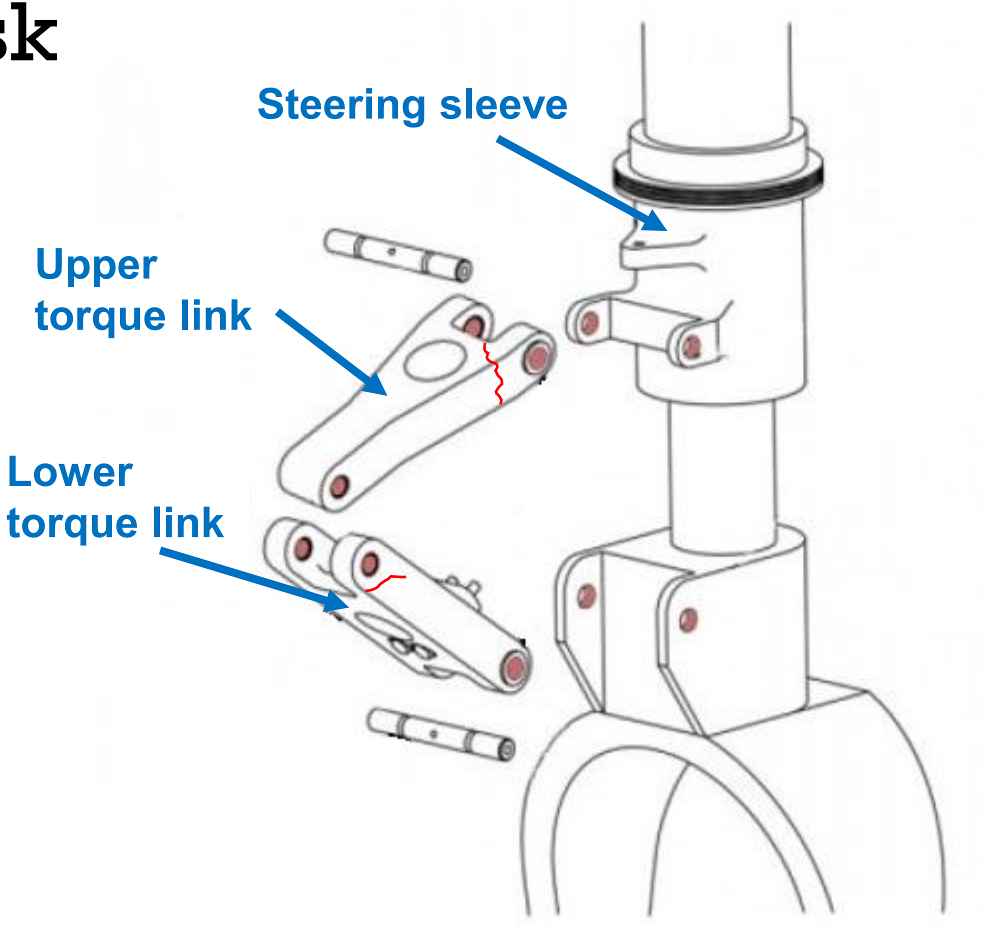
Activity	Predecessor	Duration (weeks)
A	(start)	3
B	(start)	2
C	A	8
D	B	7
E	B, C	2
F	D	2





Resource levelling task

- Torque link failed due to fatigue crack during test programme
- Your company urgently needs to *design*, *manufacture**, *measure and* assemble and test *3* new prototype parts
  - *Steering sleeve*
  - *Upper torque link*
  - *Lower torque link*
- *manufacturing is carried out sub-contract and doesn't require any resource
- Your boss wants to know:
  1. How long it will take to complete?
  2. What resources you require?



Rules:

- Your engineering team say they have **1 designer**, **1 metrologist** and **1 development** engineer already available. They have *suggested* that you will need to get additional short term contract staff to help ($$$).
- Only **designers** can **design**
- Only **metrologists** can **measure (the newly made parts)**
- Development engineers can **assemble parts** and complete the **proof test**
- **All manufacturing** is carried out sub-contract (outsourced) and **uses no internal resource**
- Avoid breaking up a task into sections as this is inefficient.
- Try to **group related tasks** logically on the chart

What to do:

1. Decide on the dependencies (sketch them out first?)
2. Create a Gantt chart using the excel sheet on Bb
3. Determine the minimum time in which the project can be done
4. Determine any resources needed
5. Validate or reject the engineering team's request for more staff
6. Upload a screengrab of your plan onto the Padlet linked in Bb. Add your group number

- Work in pairs
- Put a picture of your work on the Padlet.
- Excel file is on SharePoint week-05 Professional Practice.

List of task estimates given to you (not in order).

Task	**Duration (days)**
Manufacture new 'Upper Torque Link'	9
Measure new 'Lower Torque Link'	1
Complete proof test of final assembly	3
Measure new 'Steering sleeve'	2
Design new 'Steering sleeve'	1
Measure new 'Upper Torque Link' parts	2
Manufacture new 'Lower Torque Link'	4
Design new 'Lower Torque Link'	3
Design new 'Upper Torque Link'	3
Manufacture new 'Steering sleeve'	2
Assemble all new Parts	1

What's the normal flow for a new part?

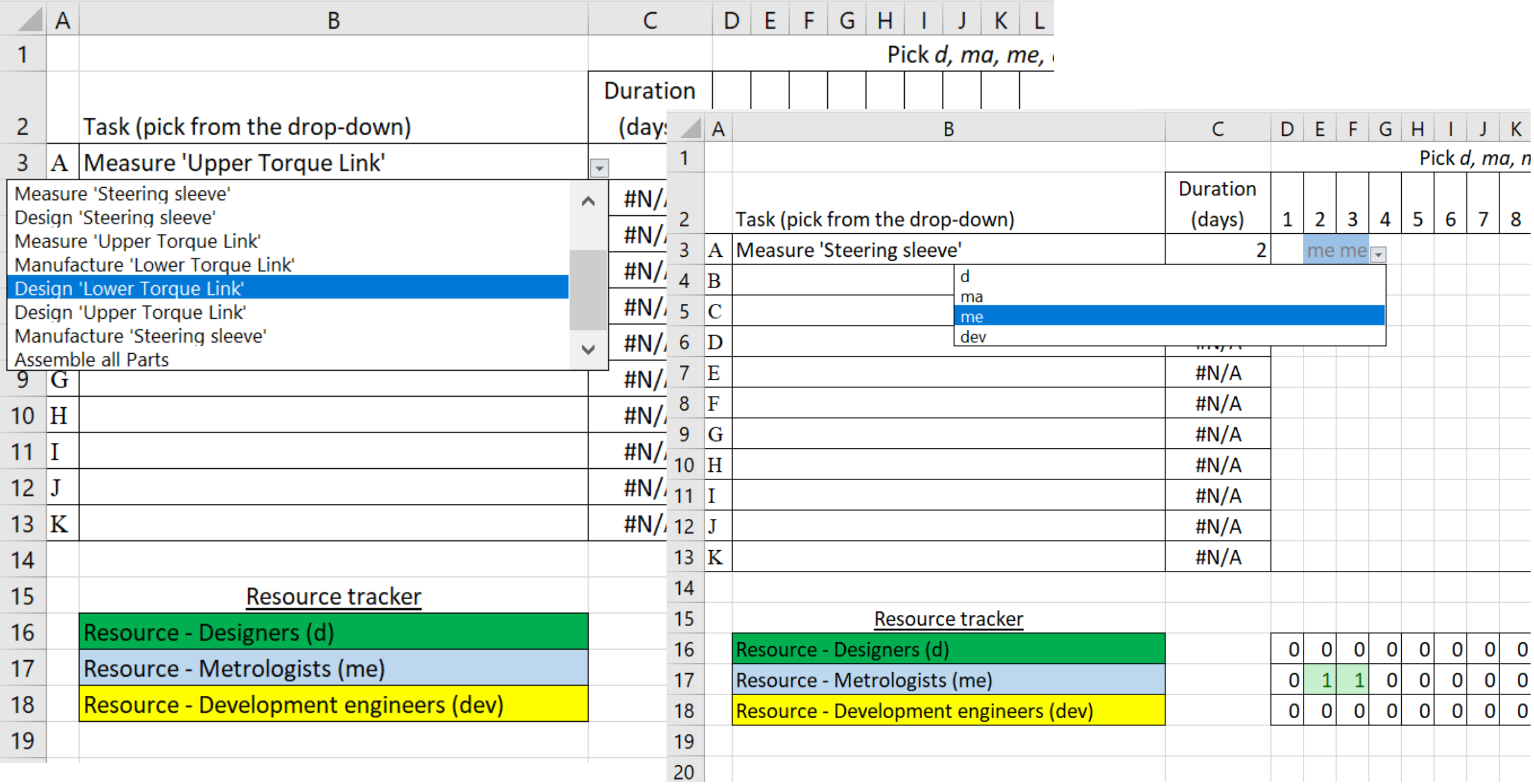
Assemble the new parts into the assembly

Test the built assembly

Measure new parts

Design new part

Manufacture new parts


Pick d, ma, me,
Task (pick from the drop-down)
Duration (days)
Measure 'Upper Torque Link'
Measure 'Steering sleeve'
Design 'Steering sleeve'
Measure 'Upper Torque Link'
Manufacture 'Lower Torque Link'
Design 'Lower Torque Link'
Design 'Upper Torque Link'
Manufacture 'Steering sleeve'
Assemble all Parts
#N/A
Resource tracker
Resource - Designers (d)
Resource - Metrologists (me)
Resource - Development engineers (dev)
Pick d, ma, n
Task (pick from the drop-down)
Duration (days)
1 2 3 4 5 6 7 8
Measure 'Steering sleeve'
2
me me
d
ma
me
dev
#N/A
Resource tracker
Resource - Designers (d)
0 0 0 0 0 0 0 0
Resource - Metrologists (me)
0 1 1 0 0 0 0 0
Resource - Development engineers (dev)
0 0 0 0 0 0 0 0

[Master file]

Too long

	Task (pick from the drop-down)	Duration (days)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
			Pick *d, ma, me, dev* from the drop-down to show the activity type																								
A	Design 'Steering sleeve'	1	d																								
B	Manufacture 'Steering sleeve'	2		ma	ma																						
C	Measure 'Steering sleeve'	2				me	me																				
D	Design 'Lower Torque Link'	3		d	d	d																					
E	Manufacture 'Lower Torque Link'	4					ma	ma	ma	ma																	
F	Measure 'Lower Torque Link'	1									me																
G	Design 'Upper Torque Link'	3					d	d	d																		
H	Manufacture 'Upper Torque Link'	9								ma	ma	ma	ma	ma	ma	ma	ma	ma									
I	Measure 'Upper Torque Link'	2																	me	me							
J	Assemble all Parts	1																			dev						
K	Complete proof test of assembly	3																				dev	dev	dev			

Resource tracker

Resource	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Resource - Designers (d)	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Resource - Metrologists (me)	0	0	0	1	1	0	0	0	1	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0
Resource - Development engineers (dev)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0

Duration 22 days ☹. No extra staff needed ☺.

Do everything as early as possible method

			Pick *d, ma, me, dev* from the drop-down to show the activity type																							
	Task (pick from the drop-down)	Duration (days)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	Design 'Upper Torque Link'	3	d	d	d																						
B	Manufacture 'Upper Torque Link'	9				ma	ma	ma	ma	ma	ma	ma	ma	ma													
C	Measure 'Upper Torque Link'	2													me	me											
D	Design 'Lower Torque Link'	3	d	d	d																						
E	Manufacture 'Lower Torque Link'	4				ma	ma	ma	ma																		
F	Measure 'Lower Torque Link'	1								me																	
G	Design 'Steering sleeve'	1	d																								
H	Manufacture 'Steering sleeve'	2		ma	ma																						
I	Measure 'Steering sleeve'	2				me	me																				
J	Assemble all Parts	1															dev										
K	Complete proof test of assembly	3																dev	dev	dev							

	Resource tracker																									
	Resource - Designers (d)		3	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	Resource - Metrologists (me)		0	0	0	1	1	0	0	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0
	Resource - Development engineers (dev)		0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0

Duration 18 days ☺. Extra design staff needed ☹.

Use the float to permit *resource levelling*: pass 1

Pick *d, ma, me, dev* from the drop-down to show the activity type

	Task (pick from the drop-down)	Duration (days)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	Design 'Upper Torque Link'	3	d	d	d																						
B	Manufacture 'Upper Torque Link'	9				ma	ma	ma	ma	ma	ma	ma	ma	ma													
C	Measure 'Upper Torque Link'	2													me	me											
D	Design 'Lower Torque Link'	3				d	d	d																			
E	Manufacture 'Lower Torque Link'	4							ma	ma	ma	ma															
F	Measure 'Lower Torque Link'	1											me														
G	Design 'Steering sleeve'	1							d																		
H	Manufacture 'Steering sleeve'	2								ma	ma																
I	Measure 'Steering sleeve'	2										me	me														
J	Assemble all Parts	1															dev										
K	Complete proof test of assembly	3																dev	dev	dev							

<u>Resource tracker</u>

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Resource - Designers (d)	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Resource - Metrologists (me)	0	0	0	0	0	0	0	0	0	1	2	0	1	1	0	0	0	0	0	0	0	0	0	0	0
Resource - Development engineers (dev)	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0

Duration 18 days ☺. 1 extra metrologist needed ☹.

Use the float to permit *resource levelling*: pass 2

			Pick *d, ma, me, dev* from the drop-down to show the activity type																							
	Task (pick from the drop-down)	Duration (days)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
A	Design 'Upper Torque Link'	3	d	d	d																						
B	Manufacture 'Upper Torque Link'	9				ma	ma	ma	ma	ma	ma	ma	ma	ma													
C	Measure 'Upper Torque Link'	2													me	me											
D	Design 'Lower Torque Link'	3				d	d	d																			
E	Manufacture 'Lower Torque Link'	4							ma	ma	ma	ma															
F	Measure 'Lower Torque Link'	1											me														
G	Design 'Steering sleeve'	1							d																		
H	Manufacture 'Steering sleeve'	2								ma	ma																
I	Measure 'Steering sleeve'	2										me	me														
J	Assemble all Parts	1															dev										
K	Complete proof test of assembly	3																de	de	dev							
	Resource tracker																										
	Resource - Designers (d)		1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0					
	Resource - Metrologists (me)		0	0	0	0	0	0	0	0	0	1	1	1	1	1	0	0	0	0	0	0					
	Resource - Development engineers (dev)		0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0					

Duration 18 days. No additional resource.

Deterrents to planning

I don’t have enough detail to plan

Start planning anyway

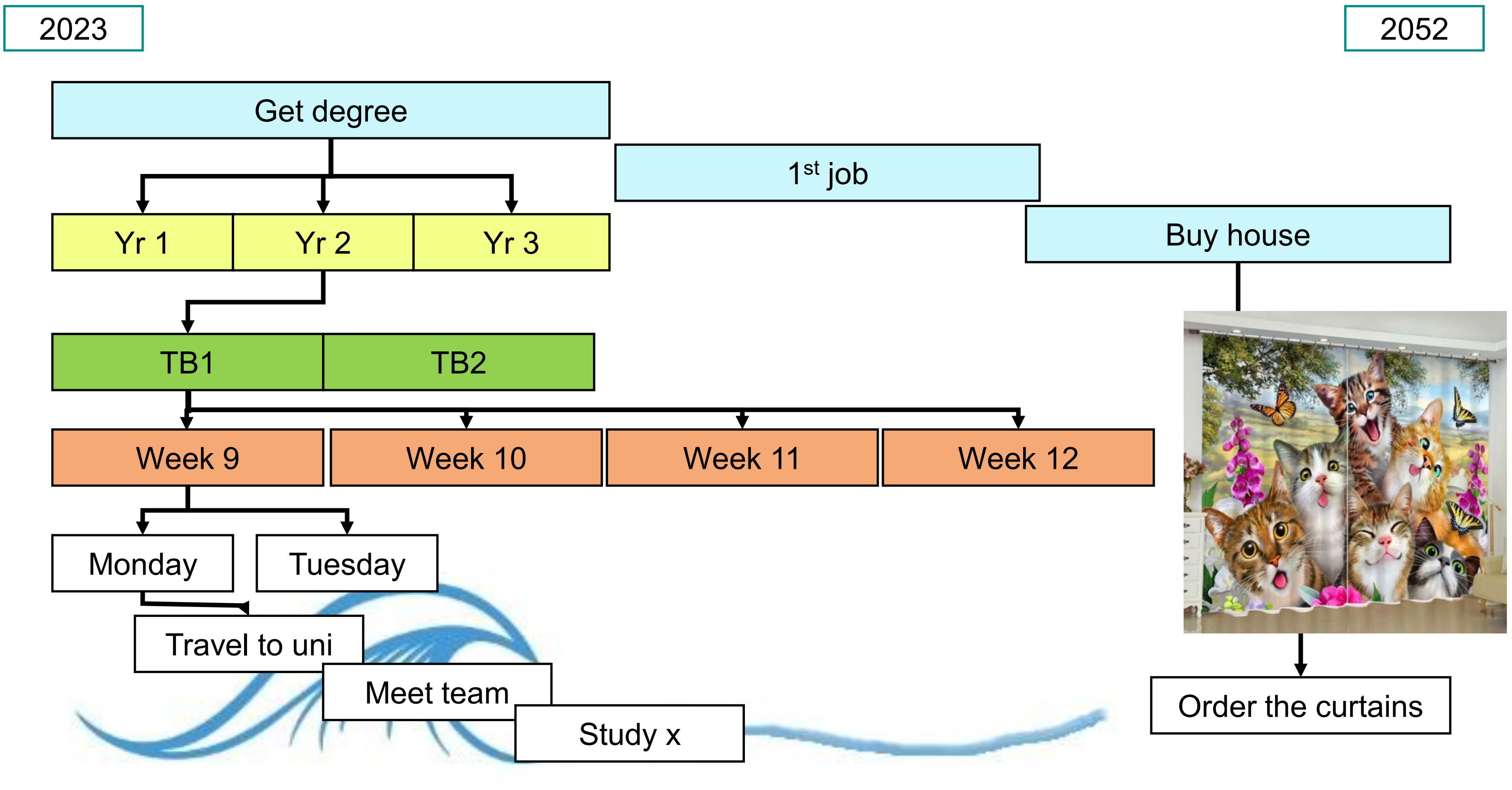
I have no idea about the durations of tasks

That’s normal.
Just guesstimate.

I only know any detail about the immediate tasks

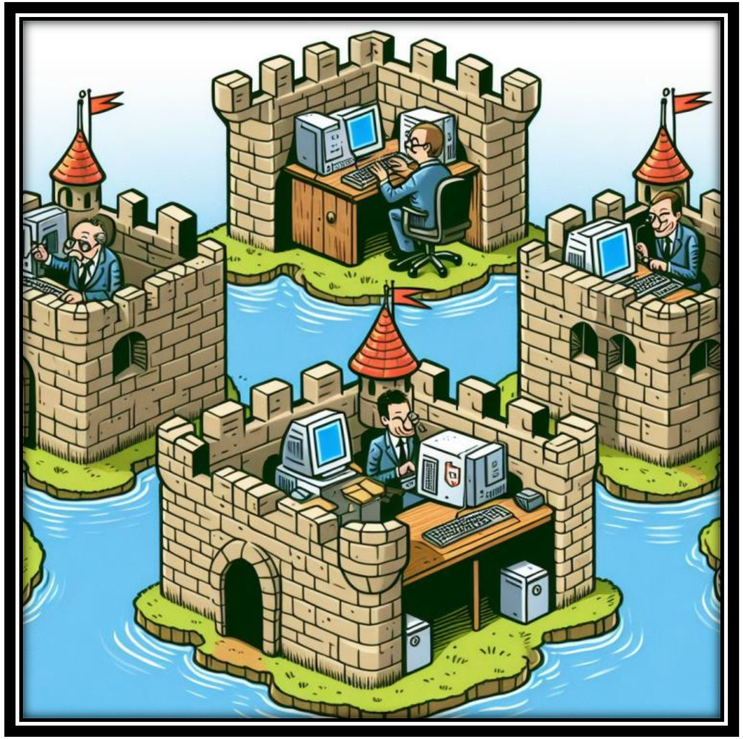
Of course - that’s called *rolling wave planning*

You already use rolling wave planning



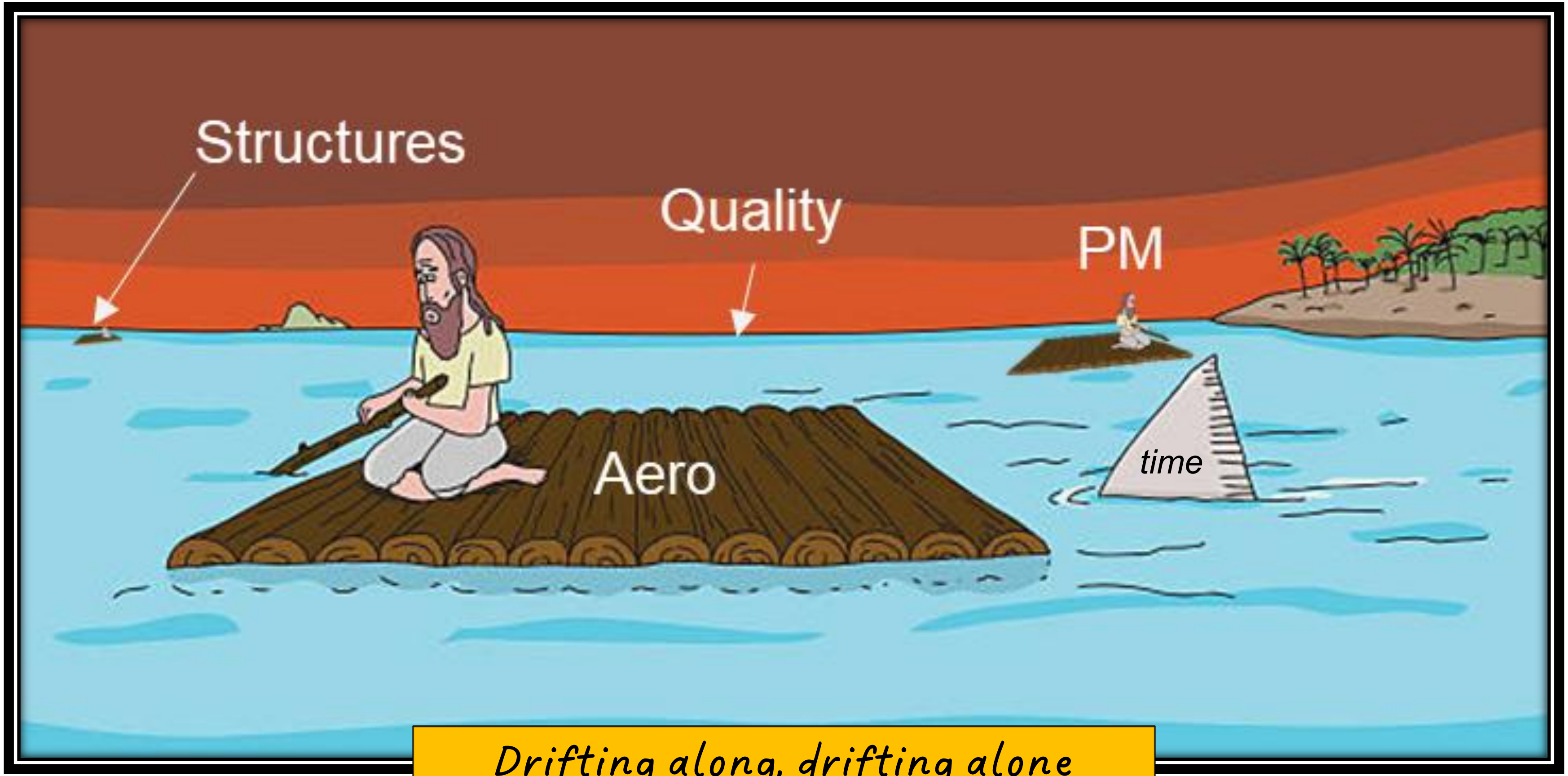
Welcome to the Museum of mediocrity

A group of individuals ≠ a team



‘Lone Genius’

A group of individuals ≠ a team



Drifting along, drifting alone

Delaying planning and design work

'I'll start when I have all the information'